Code No: 157EP **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2022
ENGINEERING MATERIALS
(Common to CE, EEE, ME, ECE, CSE)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
- - -

1.a) What are dual phase steels. Give examples and their applications.
b) Classify TRIP steels. Write down their composition, properties and uses. [7+8]

2.a) Describe HSLA steels with respect to composition , properties and applications.
b) Compare maraging and HSS steels. [8+7]

3. Compare alloys and super alloys. Briefly describe cobalt base super alloys. [15]

4. Explain the role of alloying elements in nickel base super alloys. Discuss about inconels. [15]

5. Classify the different electric materials. With help of figures describe the working of semiconductors. [15]

6.a) What are super conducting materials and their properties? Classify them and list out their uses.
b) Give examples of pyro and piezo electric materials and their applications. [8+7]

7. Compare and contrast the ferritic and austenitic stainless steels with respect to types, composition, properties and uses. [15]

8. Explain what is biocompactability. With two examples describe the applications of bio materials. [15]

---oo0oo---

JNTUH Used Papers